

M1 Understanding probability

Name: _____

Class: _____

Date: _____

Time: **413 minutes**

Marks: **340 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

The table below shows the probability distribution for a discrete random variable X .

x	0	1	2	3	4 or more
$P(X = x)$	0.35	0.25	k	0.14	0.1

Find the value of k .

Circle your answer.

0.14
0.16
0.18
1

(Total 1 mark)

Q2.

Nicola, a darts player, is practising hitting the bullseye. She knows from previous experience that she has a probability of 0.3 of hitting the bullseye with each dart.

Nicola throws eight practice darts.

- (a) Using a binomial distribution, calculate the probability that she will hit the bullseye three or more times.

(2)

- (b) Nicola throws eight practice darts on three different occasions. Calculate the probability that she will hit the bullseye three or more times on all three occasions.

(2)

- (c) State two assumptions that are necessary for the distribution you have used in part (a) to be valid.

(2)

(Total 6 marks)

Q3.

The table below shows the probability distribution for a discrete random variable X .

x	1	2	3	4	5
$P(X = x)$	k	$2k$	$4k$	$2k$	k

Find the value of k .

Circle your answer.

$\frac{1}{2}$
 $\frac{1}{4}$
 $\frac{1}{10}$
1

(Total 1 mark)

Q4.

A teacher in a college asks her mathematics students what other subjects they are studying.

She finds that, of her 24 students:

- 12 study physics
- 8 study geography
- 4 study geography and physics

- (a) A student is chosen at random from the class.

Determine whether the event 'the student studies physics' and the event 'the student studies geography' are independent.

(2)

- (b) It is known that for the whole college:

the probability of a student studying mathematics is $\frac{1}{5}$

the probability of a student studying biology is $\frac{1}{6}$

the probability of a student studying biology given that they study mathematics is $\frac{3}{8}$

Calculate the probability that a student studies mathematics or biology or both.

(4)

(Total 6 marks)

Q5.

A school took 225 children on a trip to a theme park.
After the trip the children had to write about their favourite ride at the park from a choice of three.

The table shows the number of children who wrote about each ride.

		Ride written about			Total
		The Drop	The Beanstalk	The Giant	
Year group	Year 7	24	45	23	92
	Year 8	36	17	22	75
	Year 9	20	13	25	58
Total		80	75	70	225

Three children were randomly selected from those who went on the trip.

Calculate the probability that one wrote about 'The Drop', one wrote about 'The Beanstalk'

and one wrote about 'The Giant'.

(Total 2 marks)

Q6.

A sample of 200 households was obtained from a small town.

Each household was asked to complete a questionnaire about their purchases of takeaway food.

A is the event that a household regularly purchases Indian takeaway food.

B is the event that a household regularly purchases Chinese takeaway food.

It was observed that $P(B|A) = 0.25$ and $P(A|B) = 0.1$

Of these households, 122 indicated that they did **not** regularly purchase Indian or Chinese takeaway food.

A household is selected at random from those in the sample.

Find the probability that the household regularly purchases **both** Indian and Chinese takeaway food.

(Total 6 marks)

Q7.

Roger is an active retired lecturer. Each day after breakfast, he decides whether the weather for that day is going to be fine (F), dull (D) or wet (W). He then decides on only one of four activities for the day: cycling (C), gardening (G), shopping (S) or relaxing (R). His decisions from day to day may be assumed to be independent.

The table shows Roger's probabilities for each combination of weather and activity.

		Weather		
		Fine (F)	Dull (D)	Wet (W)
Activity	Cycling (C)	0.30	0.10	0
	Gardening (G)	0.25	0.05	0
	Shopping (S)	0	0.10	0.05
	Relaxing (R)	0	0.05	0.10

- (a) Find the probability that, on a particular day, Roger decided:
- (i) that it was going to be fine and that he would go cycling;
 - (ii) on either gardening or shopping;
 - (iii) to go cycling, given that he had decided that it was going to be fine;
 - (iv) **not** to relax, given that he had decided that it was going to be dull;

(v) that it was going to be fine, given that he did **not** go cycling.

(9)

- (b) Calculate the probability that, on a particular Saturday and Sunday, Roger decided that it was going to be fine and decided on the same activity for both days.

(3)

(Total 12 marks)

Q8.

Alison is a member of a tenpin bowling club which meets at a bowling alley on Wednesday and Thursday evenings.

The probability that she bowls on a Wednesday evening is 0.90. Independently, the probability that she bowls on a Thursday evening is 0.95.

- (a) Calculate the probability that, during a particular week, Alison bowls on:

- (i) two evenings;
- (ii) exactly one evening.

(3)

- (b) David, a friend of Alison, is a member of the same club.

The probability that he bowls on a Wednesday evening, given that Alison bowls on that evening, is 0.80. The probability that he bowls on a Wednesday evening, given that Alison does not bowl on that evening, is 0.15.

The probability that he bowls on a Thursday evening, given that Alison bowls on that evening, is 1. The probability that he bowls on a Thursday evening, given that Alison does not bowl on that evening, is 0.

Calculate the probability that, during a particular week:

- (i) Alison and David bowl on a Wednesday evening;
- (ii) Alison and David bowl on both evenings;
- (iii) Alison, but not David, bowls on a Thursday evening;
- (iv) neither bowls on either evening.

(2)

(2)

(1)

(3)

(Total 11 marks)

Q9.

On a rail route between two stations, A and B, 90% of trains leave A on time and 10% of trains leave A late.

Of those trains that leave A on time, 15% arrive at B early, 75% arrive on time and 10%

arrive late.

Of those trains that leave A late, 35% arrive at B on time and 65% arrive late.

- (a) Represent this information by a fully-labelled tree diagram.

(3)

- (b) Hence, or otherwise, calculate the probability that a train:

- (i) arrives at B early or on time;
- (ii) left A on time, given that it arrived at B on time;
- (iii) left A late, given that it was not late in arriving at B.

(7)

- (c) Two trains arrive late at B. Assuming that their journey times are independent, calculate the probability that exactly one train left A on time.

(4)

(Total 14 marks)

Q10.

A survey of the 640 properties on an estate was undertaken. Part of the information collected related to the number of bedrooms and the number of toilets in each property.

This information is shown in the table.

		Number of toilets				Total
		1	2	3	4 or more	
Number of bedrooms	1	46	14	0	0	60
	2	24	67	23	0	114
	3	7	72	99	16	194
	4	0	19	123	48	190
	5 or more	0	0	11	71	82
Total		77	172	256	135	640

- (a) A property on the estate is selected at random.

Find, giving your answer to three decimal places, the probability that the property has:

- (i) exactly 3 bedrooms;

(1)

- (ii) at least 2 toilets; (2)
- (iii) exactly 3 bedrooms and at least 2 toilets; (2)
- (iv) at most 3 bedrooms, given that it has exactly 2 toilets. (3)
- (b) Three properties are selected at random from those on the estate which have exactly 3 bedrooms.

Calculate the probability that one property has 2 toilets, one has 3 toilets and the other has at least 4 toilets. Give your answer to three decimal places.

(4)
(Total 12 marks)

Q11.

A survey of the 640 properties on an estate was undertaken. Part of the information collected related to the number of bedrooms and the number of toilets in each property.

This information is shown in the table.

		Number of toilets				Total
		1	2	3	4 or more	
Number of bedrooms	1	46	14	0	0	60
	2	24	67	23	0	114
	3	7	72	99	16	194
	4	0	19	123	48	190
	5 or more	0	0	11	71	82
Total		77	172	256	135	640

- (a) A property on the estate is selected at random.
- Find, giving your answer to three decimal places, the probability that the property has:
- (i) exactly 3 bedrooms; (1)
- (ii) at least 2 toilets; (2)

- (iii) exactly 3 bedrooms and at least 2 toilets; (2)
- (iv) at most 3 bedrooms, given that it has exactly 2 toilets. (3)
- (b) Use relevant answers from part (a) to show that the number of toilets is **not** independent of the number of bedrooms. (2)
- (c) Three properties are selected at random from those on the estate which have exactly 3 bedrooms.
- Calculate the probability that one property has 2 toilets, one has 3 toilets and the other has at least 4 toilets. Give your answer to three decimal places. (4)
- (Total 14 marks)

Q12.

The number of MPs in the House of Commons was 645 at the beginning of August 2009. The genders of these MPs and the political parties to which they belonged are shown in the table.

		Political Party				Total
		Labour	Conservative	Liberal Democrat	Other	
Gender	Male	255	175	54	35	519
	Female	94	18	9	5	126
Total		349	193	63	40	645

- (a) One MP was selected at random for an interview. Calculate, to three decimal places, the probability that the MP was:
- (i) a male Conservative; (1)
- (ii) a male; (1)
- (iii) a Liberal Democrat; (1)
- (iv) Labour, given that the MP was female. (2)
- (b) Two female MPs were selected at random for an enquiry. Calculate, to three decimal places, the probability that both MPs were Labour.

(2)

- (c) Three MPs were selected at random for a committee. Calculate, to three decimal places, the probability that exactly one MP was Labour and exactly one MP was Conservative.

(4)

(Total 11 marks)

Q13.

The number of MPs in the House of Commons was 645 at the beginning of August 2009. The genders of these MPs and the political parties to which they belonged are shown in the table.

		Political Party				
		Labour	Conservative	Liberal Democrat	Other	Total
Gender	Male	255	175	54	35	519
	Female	94	18	9	5	126
	Total	349	193	63	40	645

- (a) One MP was selected at random for an interview. Calculate, to three decimal places, the probability that the MP was:
- (i) a male Conservative; (1)
 - (ii) a male; (1)
 - (iii) a Liberal Democrat; (1)
 - (iv) Labour, given that the MP was female; (2)
 - (v) male, given that the MP was not Labour. (3)
- (b) Two female MPs were selected at random for an enquiry. Calculate, to three decimal places, the probability that both MPs were Labour. (2)
- (c) Three MPs were selected at random for a committee. Calculate, to three decimal places, the probability that exactly one MP was Labour and exactly one MP was Conservative. (4)

(Total 14 marks)

Q14.

Clay pigeon shooting is the sport of shooting at special flying clay targets with a shotgun.

- (a) Rhys, a novice, uses a single-barrelled shotgun. The probability that he hits a target is 0.45, and may be assumed to be independent from target to target.

Determine the probability that, in a series of shots at 15 targets, he hits:

- (i) at most 5 targets; (1)
- (ii) more than 10 targets; (2)
- (iii) exactly 6 targets; (2)
- (iv) at least 5 but at most 10 targets. (3)

- (b) Sasha, an expert, uses a double-barrelled shotgun. She shoots at each target with the gun's first barrel and, only if she misses, does she then shoot at the target with the gun's second barrel.

The probability that she hits a target with a shot using her gun's first barrel is 0.85. The conditional probability that she hits a target with a shot using her gun's second barrel, given that she has missed the target with a shot using her gun's first barrel, is 0.80. Assume that Sasha's shooting is independent from target to target.

- (i) Show that the probability that Sasha hits a target is 0.97. (2)
- (ii) Determine the probability that, in a series of shots at 50 targets, Sasha hits at least 48 targets. (3)
- (iii) In a series of shots at 80 targets, calculate the mean number of times that Sasha shoots at targets with her gun's second barrel. (2)

(Total 15 marks)

Q15.

Emma visits her local supermarket every Thursday to do her weekly shopping.

The event that she buys orange juice is denoted by J , and the event that she buys bottled water is denoted by W . At each visit, Emma may buy neither, or one, or both of these items.

- (a) Complete the table of probabilities, printed below, for these events, where J' and W' denote the events 'not J ' and 'not W ' respectively. (3)

- (b) Hence, or otherwise, find the probability that, on any given Thursday, Emma buys either orange juice or bottled water but not both.

(2)

- (c) Show that:

- (i) the events J and W are **not** mutually exclusive;
- (ii) the events J and W are **not** independent.

	J	J'	Total
W			0.65
W'	0.15		
Total		0.30	1.00

(3)

(Total 8 marks)

Q16.

- (a) Emma visits her local supermarket every Thursday to do her weekly shopping.

The event that she buys orange juice is denoted by J , and the event that she buys bottled water is denoted by W . At each visit, Emma may buy neither, or one, or both of these items.

- (i) Complete the table of probabilities, printed below, for these events, where J' and W' denote the events 'not J ' and 'not W ' respectively.

(3)

- (ii) Hence, or otherwise, find the probability that, on any given Thursday, Emma buys either orange juice or bottled water but not both.

(2)

- (iii) Show that:

(A) the events J and W are **not** mutually exclusive;

(B) the events J and W are **not** independent.

(3)

- (b) Rhys visits the supermarket every Saturday to do his weekly shopping. Items that he may buy are milk, cheese and yogurt.

The probability, $P(M)$, that he buys milk on any given Saturday is 0.85.

The probability, $P(C)$, that he buys cheese on any given Saturday is 0.60.

The probability, $P(Y)$, that he buys yogurt on any given Saturday is 0.55.

The events M , C and Y may be assumed to be independent.

Calculate the probability that, on any given Saturday, Rhys buys:

- (i) none of the 3 items;

(2)

- (ii) exactly 2 of the 3 items.

	J	J'	Total
W			0.65
W'	0.15		
Total		0.30	1.00

(3)

(Total 13 marks)

Q17.

An IT help desk has three telephone stations: Alpha, Beta and Gamma. Each of these stations deals only with telephone enquiries.

The probability that an enquiry is received at Alpha is 0.60.

The probability that an enquiry is received at Beta is 0.25.

The probability that an enquiry is received at Gamma is 0.15.

Each enquiry is resolved at the station that receives the enquiry. The **percentages** of enquiries resolved within various times at **each** station are shown in the table.

		Time		
		≤ 1 hour	≤ 24 hours	≤ 72 hours
Station	Alpha	55	80	100
	Beta	60	85	100
	Gamma	40	75	100

For example:

80 per cent of enquiries received at Alpha are resolved within 24 hours;

25 per cent of enquiries received at Alpha take between 1 hour and 24 hours to resolve.

- (a) Find the probability that an enquiry, selected at random, is:

- (i) resolved at Gamma;

(1)

- (ii) resolved at Alpha within 1 hour;

(1)

- (iii) resolved within 24 hours;

(2)

- (iv) received at Beta, given that it is resolved within 24 hours.

(3)

- (b) A random sample of 3 enquiries was selected.

Given that all 3 enquiries were resolved within 24 hours, calculate the probability that they were all received at:

- (i) Beta;

(2)

- (ii) the same station.

(4)

(Total 13 marks)

Q18.

Each school-day morning, three students, Rita, Said and Ting, travel independently from their homes to the same school by one of three methods: walk, cycle or bus. The table shows the probabilities of their independent daily choices.

	Walk	Cycle	Bus
Rita	0.65	0.10	0.25
Said	0.40	0.45	0.15
Ting	0.25	0.55	0.20

- (a) Calculate the probability that, on any given school-day morning:

- (i) all 3 students walk to school;

(2)

- (ii) only Rita travels by bus to school;

(2)

- (iii) at least 2 of the 3 students cycle to school.

(4)

- (b) Ursula, a friend of Rita, never travels to school by bus. The probability that:

Ursula walks to school when Rita walks to school is 0.9;
 Ursula cycles to school when Rita cycles to school is 0.7.

Calculate the probability that, on any given school-day morning, Rita and Ursula travel to school by:

- (i) the same method;

(3)

- (ii) different methods.

(1)

(Total 12 marks)

Q19.

Gary and his neighbour Larry work at the same place.

On any day when Gary travels to work, he uses one of three options: his car only, a bus only or both his car and a bus. The probability that he uses his car, either on its own or with a bus, is 0.6. The probability that he uses both his car and a bus is 0.25.

(a) Calculate the probability that, on any particular day when Gary travels to work, he:

(i) does not use his car;

(1)

(ii) uses his car only;

(2)

(iii) uses a bus.

(3)

(b) On any day, the probability that Larry travels to work with Gary is 0.9 when Gary uses his car only, is 0.7 when Gary uses both his car and a bus, and is 0.3 when Gary uses a bus only.

Calculate the probability that, on any particular day when Gary travels to work, Larry travels with him.

(4)

(Total 10 marks)

Q20.

Gary and his neighbour Larry work at the same place.

On any day when Gary travels to work, he uses one of three options: his car only, a bus only or both his car and a bus. The probability that he uses his car, either on its own or with a bus, is 0.6. The probability that he uses both his car and a bus is 0.25.

(a) Calculate the probability that, on any particular day when Gary travels to work, he:

(i) does not use his car;

(1)

(ii) uses his car only;

(2)

(iii) uses a bus.

(3)

(b) On any day, the probability that Larry travels to work with Gary is 0.9 when Gary uses his car only, is 0.7 when Gary uses both his car and a bus, and is 0.3 when Gary uses a bus only.

(i) Calculate the probability that, on any particular day when Gary travels to work, Larry travels with him.

(4)

- (ii) Assuming that option choices are independent from day to day, calculate, to three decimal places, the probability that, during any particular week (5 days) when Gary travels to work every day, Larry never travels with him.

(2)

(Total 12 marks)

Q21.

A large bookcase contains two types of book: hardback and paperback. The number of books of each type in each of four subject categories is shown in the table.

		Subject category				Total
		Crime	Romance	Science fiction	Thriller	
Type	Hardback	8	16	18	18	60
	Paperback	16	40	14	30	100
Total		24	56	32	48	160

- (a) A book is selected at random from the bookcase. Calculate the probability that the book is:

(i) a paperback;

(1)

(ii) not science fiction;

(2)

(iii) science fiction or a hardback;

(2)

(iv) a thriller, given that it is a paperback.

(2)

- (b) Three books are selected at random, without replacement, from the bookcase.

Calculate, to three decimal places, the probability that one is crime, one is romance and one is science fiction.

(4)

(Total 11 marks)

Q22.

David and his partner, Amber, both work at their district hospital. The probability that David cycles to work each morning is 0.85. On a morning when David cycles to work, the probability that Amber cycles to work is 0.80. On a morning when David does not cycle to work, the probability that Amber cycles to work is 0.30.

- (a) Calculate the probability that, on a particular morning:

- (i) they both cycle to work; (2)
 - (ii) only Amber cycles to work; (2)
 - (iii) exactly one of them cycles to work. (2)
 - (b) Assuming that decisions as to whether to cycle to work are independent from day to day, calculate the probability that, during a period of 4 working days, they both cycle to work on exactly 3 of the days and neither of them cycles to work on the other day. (4)
- (Total 10 marks)**

Q23.

A health club has a number of facilities which include a gym and a sauna. Andrew and his wife, Heidi, visit the health club together on Tuesday evenings.

On any visit, Andrew uses either the gym or the sauna or both, but no other facilities. The probability that he uses the gym, $P(G)$, is 0.70. The probability that he uses the sauna, $P(S)$, is 0.55. The probability that he uses both the gym and the sauna is 0.25.

- (a) Calculate the probability that, on a particular visit:
 - (i) he does not use the gym; (1)
 - (ii) he uses the gym but not the sauna; (2)
 - (iii) he uses either the gym or the sauna but not both. (2)
- (b) Assuming that Andrew's decision on what facility to use is independent from visit to visit, calculate the probability that, during a month in which there are exactly four Tuesdays, he does not use the gym. (2)
- (c) The probability that Heidi uses the gym when Andrew uses the gym is 0.6, but is only 0.1 when he does not use the gym.
 Calculate the probability that, on a particular visit, Heidi uses the gym. (3)
- (d) On any visit, Heidi uses **exactly one** of the club's facilities.
 The probability that she uses the sauna is 0.35.
 Calculate the probability that, on a particular visit, she uses a facility other than the gym or the sauna. (2)

(Total 12 marks)

Q24.

A travel agency in Tunisia offers customers a 3-day tour into the Sahara desert by either coach or minibus.

- (a) The agency accepts bookings from 50 customers for seats on the coach. The probability that a customer, who has booked a seat on the coach, will **not** turn up to claim the seat is 0.08, and may be assumed to be independent of the behaviour of other customers.

Determine the probability that, of the customers who have booked a seat on the coach:

- (i) two or more will **not** turn up;
(ii) three or more will **not** turn up.

(4)

- (b) The agency accepts bookings from 15 customers for seats on the minibus. The probability that a customer, who has booked a seat on the minibus, will **not** turn up to claim the seat is 0.025, and may be assumed to be independent of the behaviour of other customers.

Calculate the probability that, of the customers who have booked a seat on the minibus:

- (i) all will turn up;
(ii) one or more will **not** turn up.

(4)

- (c) The coach has 48 seats and the minibus has 14 seats. If 14 or fewer customers who have booked seats on the minibus turn up, they will be allocated a seat on the minibus. If all 15 customers who have booked seats on the minibus turn up, one will be allocated a seat on the coach. This will leave only 47 seats available for the 50 customers who have booked seats on the coach.

Use your results from parts (a) and (b) to calculate the probability that there will be seats available on the coach for all those who turn up having booked such seats.

(4)

(Total 12 marks)

Q25.

Rea, Suki and Tora take part in a shooting competition. The final round of the competition requires each of them to try to hit the centre of a target, placed at 100 metres, with a single shot. The independent probabilities that Rea, Suki and Tora hit the centre of this target with a single shot are 0.7, 0.6 and 0.8 respectively.

Find the probability that, in the final round of the competition, the centre of the target will be hit by:

- (a) Tora only; (2)
 - (b) exactly one of the three competitors; (3)
 - (c) at least one of the three competitors. (3)
- (Total 8 marks)**

Q26.

Dafydd, Eli and Fabio are members of an amateur cycling club that holds a time trial each Sunday during the summer. The independent probabilities that Dafydd, Eli and Fabio take part in any one of these trials are 0.6, 0.7 and 0.8 respectively.

Find the probability that, on a particular Sunday during the summer:

- (a) none of the three cyclists takes part; (2)
 - (b) Fabio is the only one of the three cyclists to take part; (2)
 - (c) exactly one of the three cyclists takes part; (3)
 - (d) either one or two of the three cyclists take part. (3)
- (Total 10 marks)**

Q27.

Xavier, Yuri and Zara attend a sports centre for their judo club's practice sessions. The probabilities of them arriving late are, independently, 0.3, 0.4 and 0.2 respectively.

- (a) Calculate the probability that for a particular practice session:
 - (i) all three arrive late; (1)
 - (ii) none of the three arrives late; (2)
 - (iii) only Zara arrives late. (2)
- (b) Zara's friend, Wei, also attends the club's practice sessions. The probability that Wei arrives late is 0.9 when Zara arrives late, and is 0.25 when Zara does not arrive late.

Calculate the probability that for a particular practice session:
 - (i) both Zara and Wei arrive late;

(2)

- (ii) either Zara or Wei, but not both, arrives late.

(3)

(Total 10 marks)

Q28.

Each enquiry received by a business support unit is dealt with by Ewan, Fay or Gaby. The probabilities of them dealing with an enquiry are 0.2, 0.3 and 0.5 respectively.

Of enquiries dealt with by Ewan, 60% are answered immediately, 25% are answered later the same day and the remainder are answered at a later date.

Of enquiries dealt with by Fay, 75% are answered immediately, 15% are answered later the same day and the remainder are answered at a later date.

Of enquiries dealt with by Gaby, 90% are answered immediately and the remainder are answered at a later date.

- (a) Determine the probability that an enquiry:

- (i) is dealt with by Gaby and answered immediately;

(1)

- (ii) is answered immediately;

(3)

- (iii) is dealt with by Gaby, given that it is answered immediately.

(3)

- (b) Determine the probability that an enquiry is dealt with by Ewan, given that it is answered later the same day.

(4)

(Total 11 marks)

Q29.

A reliable estimate for the proportion of a population of fish with a certain disease is 60 per cent.

A test for the presence of the disease in a fish is possible. The test gives one of three conclusions: diseased, inconclusive, non-diseased.

For a **diseased** fish, the probabilities of these three conclusions are:

diseased	0.75
inconclusive	0.15
non-diseased	0.10

For a **non-diseased** fish, the probabilities of these three conclusions are:

diseased	0.05
inconclusive	0.15

non-diseased 0.80

- (a) A fish is selected at random. Using a tree diagram, or otherwise, calculate the probability that:

(i) the fish has the disease and the test concludes that it is diseased; (2)

(ii) the test concludes that the fish has the disease; (3)

(iii) the test gives a correct conclusion. (2)

- (b) Three fish, all with the disease, are tested.

Find the probability that the test concludes that two fish are diseased and one fish is non-diseased.

(3)

(Total 10 marks)

Q30.

Fred and his daughter, Delia, support their town's rugby team. The probability that Fred watches a game is 0.8. The probability that Delia watches a game is 0.9 when her father watches the game, and is 0.4 when her father does not watch the game.

- (a) Calculate the probability that:

(i) both Fred and Delia watch a particular game; (2)

(ii) neither Fred nor Delia watch a particular game. (2)

- (b) Molly supports the same rugby team as Fred and Delia. The probability that Molly watches a game is 0.7, and is independent of whether or not Fred or Delia watches the game.

Calculate the probability that:

(i) all 3 supporters watch a particular game; (3)

(ii) exactly 2 of the 3 supporters watch a particular game. (4)

(Total 11 marks)

Q31.

Of the patients in an accident unit, 30% have a sports injury.

Past records for the accident unit suggest that:

a patient with a sports injury has a probability of 0.2 of being admitted to hospital;

a patient who does **not** have a sports injury has a probability of 0.4 of being admitted to hospital.

A patient is chosen at random from those in the accident unit.

- (a) (i) Write down the probability that the chosen patient has a sports injury. (1)
- (ii) Find the probability that the chosen patient has a sports injury and is admitted to hospital. (1)
- (iii) Show that the probability that the chosen patient is admitted to hospital is 0.34. (2)
- (b) Given that the chosen patient is admitted to hospital, find the conditional probability that the patient has a sports injury. (3)

(Total 7 marks)

Q32.

A customer goes into a store to buy a refrigerator and a microwave. From past experience it is known that 10% of the refrigerators and 5% of the microwaves will be found to be defective when tested. The customer chooses one refrigerator and one microwave at random and the items are tested.

- (a) Find the probability that:
- (i) both items are found to be defective; (2)
- (ii) neither item is found to be defective; (2)
- (iii) exactly one of the items is found to be defective. (2)
- (b) Given that exactly one of the items is found to be defective, find the probability that it is the refrigerator. (3)

(Total 9 marks)

Q33.

A football club employs 30 players, of whom 13 have British nationality. The following table shows how many of each type of player are employed by the club.

	Goalkeepers	Defenders	Midfielders	Attackers

British	2	4	5	2
Other	2	5	7	3

One player is chosen at random from the 30 players.

- (a) Find the probability that the chosen player is:
- (i) British; (1)
 - (ii) a goalkeeper; (1)
 - (iii) British and a goalkeeper; (1)
 - (iv) a defender or a midfielder; (1)
 - (v) British or an attacker but **not** both. (2)
- (b) (i) Given that the chosen player is British, find the probability that he is a goalkeeper. (2)
- (ii) State, giving a reason, whether the events
 'the chosen player is British' and
 'the chosen player is a goalkeeper'
 are independent. (2)
- (Total 10 marks)

Q34.

Jane is an athlete in training for a major event. Every evening she does either a 5-mile run or a 10-mile run. On a given day Jane does a full day's work with probability $\frac{2}{3}$.

When she has done a full day's work, the probability that she does a 10-mile run is $\frac{2}{5}$.

When she has done a **not** full day's work, the probability that she does a 10-mile run is $\frac{4}{5}$.

- (a) For a random day, calculate the probability that:
- (i) she does a full day's work and a 10-mile run; (2)

- (ii) she does **not** do a full day's work but she does a 10-mile run; (2)
- (iii) she does a 10-mile run. (1)
- (b) Given that she does a 10-mile run, find the conditional probability that she has done a full day's work. (2)
- (Total 7 marks)

Q35.

Two letters are posted on the same day.

Letter A is sent by first-class post and has a probability of 0.9 of being delivered the next day.

Letter B is sent by second-class post and has a probability of only 0.3 of being delivered the next day.

- (a) Find the probability that:
- (i) **both** letters are delivered the next day; (2)
- (ii) **neither** letter is delivered the next day; (2)
- (iii) **at least one** of the letters is delivered the next day. (2)
- (b) Given that at least one of the letters is delivered the next day, find the probability that letter A is delivered the next day. (3)
- (Total 9 marks)
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